

Team Details

a. Team name: Team Lifeline (Lifeline OS)

b. Team leader name: Mansi Dhruv

c. Problem Statement: AI-assisted flood impact intelligence and resilient emergency-response decision support under extreme climate disruptions in rural & regional Assam (Barak River Basin / Cachar District).

Track: Climate Resilience, Disaster Response, and Public Health Equity • GitHub: https://github.com/MsMansiDhruv/lifeline-os

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TEMPLATE INSTRUCTION PRESERVED — COMPILED TO NATIVE PPTX & PDF

Brief about the idea

Evidence-Driven Emergency Response Decision Support for Flooded Road Networks and Isolated Rural Populations

- **The Challenge in Regional Basins:**
  During peak monsoon surges in Assam's Barak River basin (Cachar District), rapid inundation cuts off key culverts and bridges. Disaster managers face severe evidence fragmentation: satellite passes arrive hours apart, emergency calls conflict, and relief convoys operate on intuition without knowing which communities have crossed the clinical 60-minute 'Golden Hour' cutoff.

- **What Lifeline OS Is:**
  A deterministic geospatial decision-support platform designed to transform raw multi-source flood evidence into auditable, life-saving field actions. It fuses Copernicus Sentinel-1 SAR radar, Open-Meteo precipitation, and GloFAS river discharge with OpenStreetMap road networks.

- **How It Operates:**
  Rather than just displaying passive hazard heatmaps, Lifeline OS computes percolation severance across a 31,214-node directed graph. It identifies completely isolated population clusters, calculates detour delays to Block Primary Health Centers (BPHCs), runs k-medoids optimization to position relief boat staging pads on high-elevation nodes, and evaluates temporary Bailey bridge deployments

- **Operational Truthfulness Rule:**
  No unverified damage claims. Spatial intersections are classified strictly as 'Potential Inundation / Exposure.' Where no safe road exists, the engine returns an explicit NO_ROUTE error instead of hallucinating submerged dirt tracks. All actions require signed civil commander approval under the Disaster Management Act 2005.

Opportunities

DIFFERENTIATION

How different from existing ideas?

Legacy Flood Dashboards:

- Passive monitoring of historical water rasters
- No connection between flood shapes & road networks
- Zero awareness of clinical healthcare reachability
- Intuition-based relief dispatch leads to bottlenecks
- Cloud-heavy: fails during disaster power cuts

Lifeline OS Operational Paradigm:

- Active decision support, not passive maps
- Dynamic Dijkstra routing over 31k road nodes
- 60-min Golden Hour catchment analysis
- Deterministic boat staging & bridge ROI
- Edge-first: 100% offline via IndexedDB

SOLUTION ENGINE

How will it solve the problem?

1. Cut-Edge Percolation Slicing:

Intersects flood polygons with road vectors in <850ms, identifying exactly 1,830 severed road links.

2. Clinical Healthcare Triage:

Calculates urgent demand: dialysis patients, high-risk pregnancies, and days severed from BPHCs.

3. Pre-Positioning Staging Hubs:

Runs k-medoids to station relief boats on safe, dry high-elevation nodes adjacent to cut-off clusters.

4. Micro-Intervention ROI Solver:

Ranks candidate Bailey bridge crossings:
ROI = (Reconnected Pop + Sum delta-km) / Span.

UNIQUE VALUE

USP of the proposed solution

- **Zero Route Hallucination:**

Explicit NO_ROUTE boundaries prevent sending emergency crews onto submerged causeways and collapsed culverts.

- **7-Tier Epistemic Lineage:**

Every layer is tagged Observed, Official, Reported, Modelled, Assumed, or Simulated for full clarity.

- **Offline-First Client Store:**

Local IndexedDB cache ensures commanders maintain full spatial analysis without internet during power outages.

- **Statutory Compliance & Audit:**

Disaster Management Act 2005 compliant. Cryptographic SHA-256 state ledger for post-event judicial inquiry.

List of features offered by the solution

1. Multi-Source Environmental Ingestion

INGESTION

Automated ingest adapters for Sentinel-1 SAR (10m radar), and Open-Meteo precipitation, GloFAS river discharge ensembles, OSM road networks.

2. 7-Tier Epistemic Provenance Engine

PROVENANCE

Enforces strict Zod schema validation, 6-hour temporal TTL staleness monitoring, and tags every feature with explicit confidence boundaries.

3. Topological Percolation & Cut-Edges

SPATIAL ANALYSIS

Slices 69,663 road segments in metric UTM Zone 46N to 1,830 identify severed edges without prematurely claiming physical structural collapse.

4. Dynamic Dijkstra Routing & Isolation

ACCESSIBILITY

Re-routes 319 settlements to BPHCs on a 31k-node graph; flags isolated clusters exceeding the critical 60-minute emergency Golden Hour limit.

5. Pre-Positioning Hubs (k-Medoids)

OPTIMIZATION

Calculates optimal non-flooded, high-elevation staging pads for SDRF relief boats and mobile clinics adjacent to cut-off clusters.

6. Micro-Bridge ROI Engine

CIVIL ENGINEERING

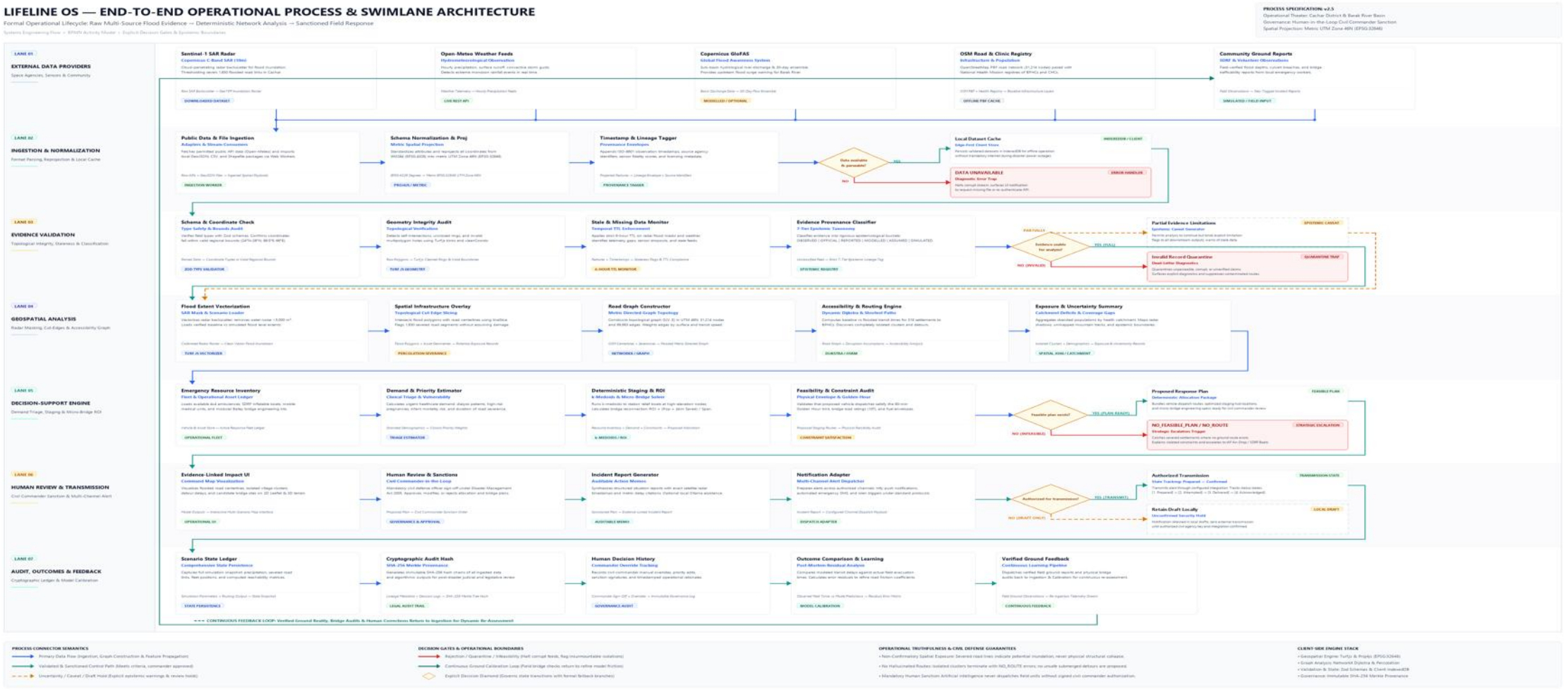
Ranks candidate modular Bailey bridge crossings and temporary pontoon bridges: ROI = (Reconnected Pop + Sum delta-km Saved) / Span Length.

7. Command Terminal & State Ledger

GOVERNANCE

Dual-panel 2D/3D tactical map, digital commander sanction order workflow, multi-channel dispatch (SMS, ntfy), and audit SHA-256 Merkle trail.

Process flow diagram or Use-case diagram



Operational Swimlanes: 1. External Providers • 2. Ingestion & Cache • 3. Epistemic Validation • 4. Spatial Graph Analysis • 5. Decision Engine • 6. Civil Commander Sanction • 7. Audit Ledger

Wireframes/Mock diagrams of the proposed solution (optional)

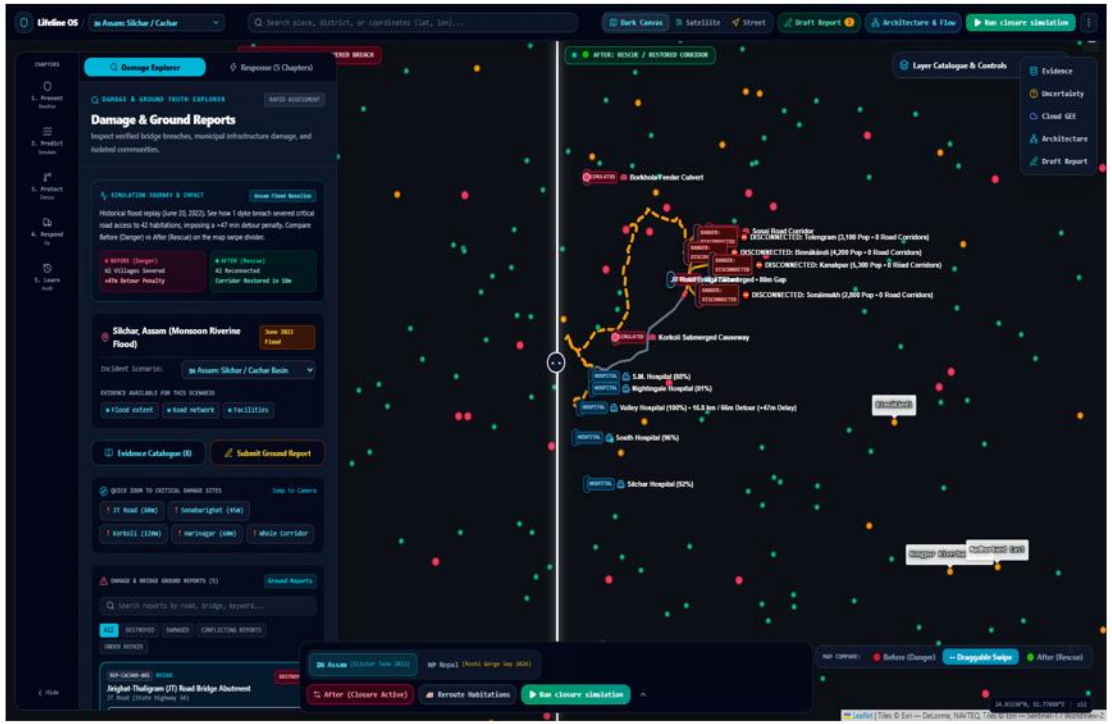


Figure 1: Dual-Panel Situational Awareness Console

- Interactive 2D Leaflet flood vectors with Esri satellite imagery
- Real-time population isolation counters and clinic accessibility badges
- Dynamic scenario selector: baseline vs moderate vs catastrophic flood levels
- Instant route severance inspection without heavy client overhead

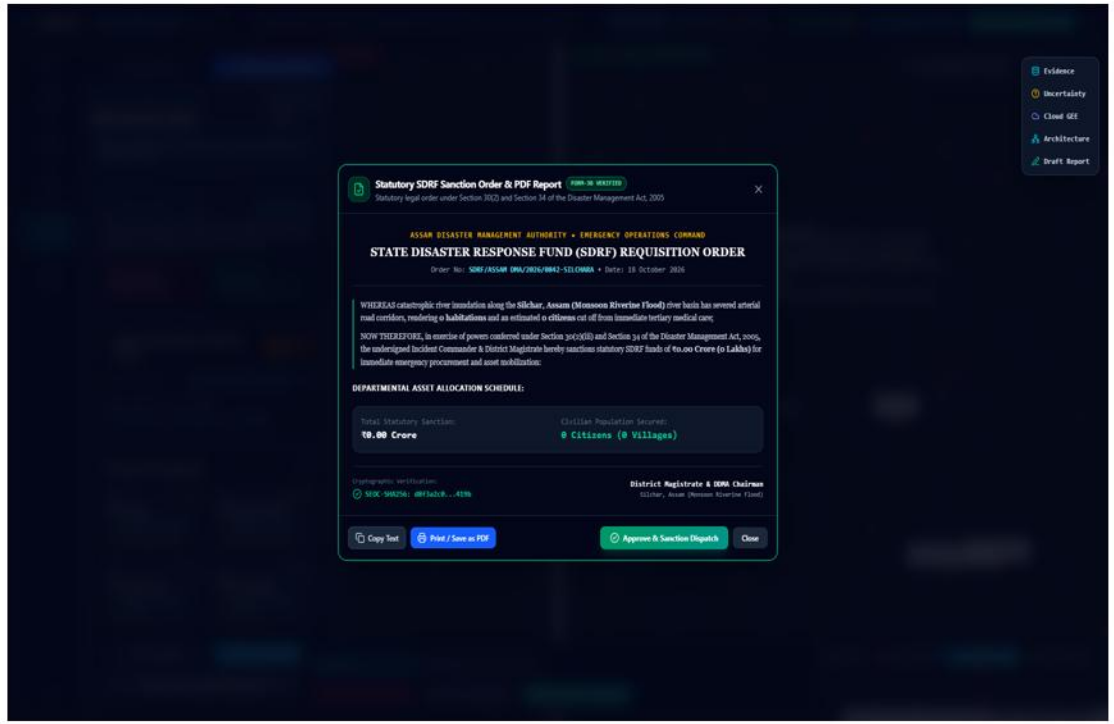
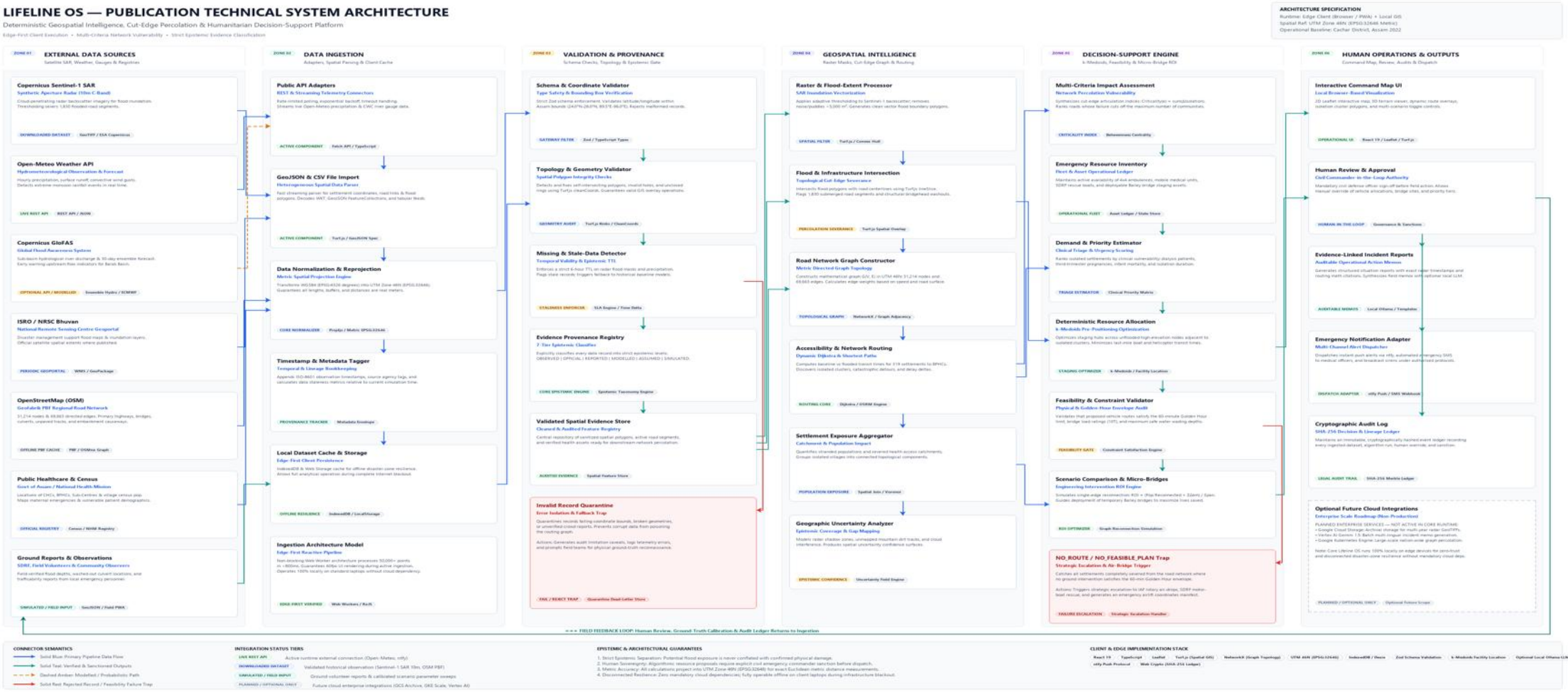


Figure 2: Commander Review & Sanction Order Terminal

- Formal statutory action memo generation with exact satellite timestamps
- Engineering specifications for candidate temporary Bailey bridge sites
- Digital signature lock complying with Disaster Management Act 2005
- Multi-channel alert dispatch (SMS, ntfy) with state confirmation logging

Architecture diagram of the proposed solution



System Layers: External Providers → Ingestion & Edge Cache → Geospatial Core (Turf.js/NetworkX) → Decision Optimization → Command Dispatch & Audit

Technologies to be used in the solution

FRONTEND & EDGE FRAMEWORK

React 19, TypeScript, Vite, Tailwind CSS, Lucide Icons, HTML5 Canvas API.

- Client-side edge Progressive Web Application (PWA) with zero cold start
- Operates seamlessly across standard field laptops and command tablets
- Fully responsive tactical split-view UI for high-stress operational rooms
- Client Web Workers handle intensive background computation without UI lag

GEOSPATIAL & NETWORK GRAPH ENGINE

Turf.js, Proj4js (EPSG:32646 UTM 46N), NetworkX, Leaflet, MapLibre GL.

- Turf.js executes vector polygon dissolves, cleanCoords, and lineSlice
- Proj4js guarantees metric distance accuracy across Cachar UTM Zone 46N
- NetworkX powers dynamic Dijkstra routing and topological cut-edge percolation
- Leaflet + MapLibre render rich 2D vector overlays and 3D elevation terrain

VALIDATION, PROVENANCE & CLIENT STORAGE

Zod Schemas, IndexedDB (Dexie/idb), Web Workers, SHA-256 Merkle Chains.

- Strict Zod type safety eliminates unparseable, corrupted, or spoofed feeds
- IndexedDB edge store retains road networks & radar rasters during internet outages
- Multi-threaded Web Workers prevent browser UI freezes during 31k-node traversal
- SHA-256 Merkle hash chains guarantee legal provenance for judicial review

CLOUD, GEOSPATIAL APIS & DISPATCH SERVICES

Google Cloud Storage, Cloud Run, Firebase Firestore, Sentinel-1 API, ntfy, Twilio.

- Google Cloud Storage stores tiled GeoTIFF rasters and baseline PBF networks
- Google Cloud Run hosts lightweight serverless calibration and sync endpoints
- Firebase Firestore synchronizes real-time command orders across field teams
- Multi-channel alert dispatch adapters: ntfy push notifications and Twilio SMS

Estimated implementation cost (optional)

Transparent Public-Sector Deployment Model (Budget per District Disaster Management Authority)

- **Client-Side Architectural Advantage (Zero Server Overhead):**

Because all graph traversal and spatial calculations run client-side in browser Web Workers, field offices and District Emergency Operation Centers (DEOCs) run Lifeline OS on existing standard duty laptops (Core i5 / 8GB RAM). Zero expensive GPU clusters or dedicated cloud supercomputers required.

- **Cloud Infrastructure Run-Rate (Google Cloud Platform):**

• Google Cloud Run (Microservices API): ~$15 / month (standby); ~$45 / month (disaster surge)
• Google Cloud Storage (GeoTIFF tile cache & road vectors): ~$10 / month
• Firebase Firestore (Live multi-user command synchronization): ~$20 / month

- **Satellite Data & Telemetry Ingestion:**

• Copernicus Sentinel-1 SAR: $0 (Free & open access via Copernicus Data Space Ecosystem)
• Open-Meteo Weather: $0 (Permitted public tier) or $35/mo (commercial SLA tier)
• OpenStreetMap Infrastructure: $0 (Open Database License - ODbL).

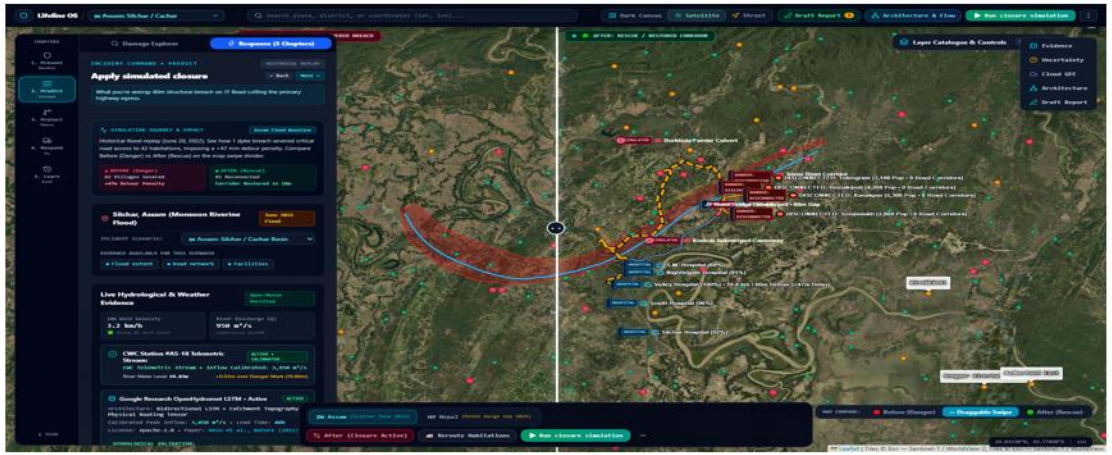
- **Annual Total Cost of Ownership (TCO):**

Estimated at <$1,200 / year per district. Compares against legacy enterprise GIS solutions costing >$40,000 in annual recurring seat licenses while lacking automated emergency accessibility and staging algorithms.

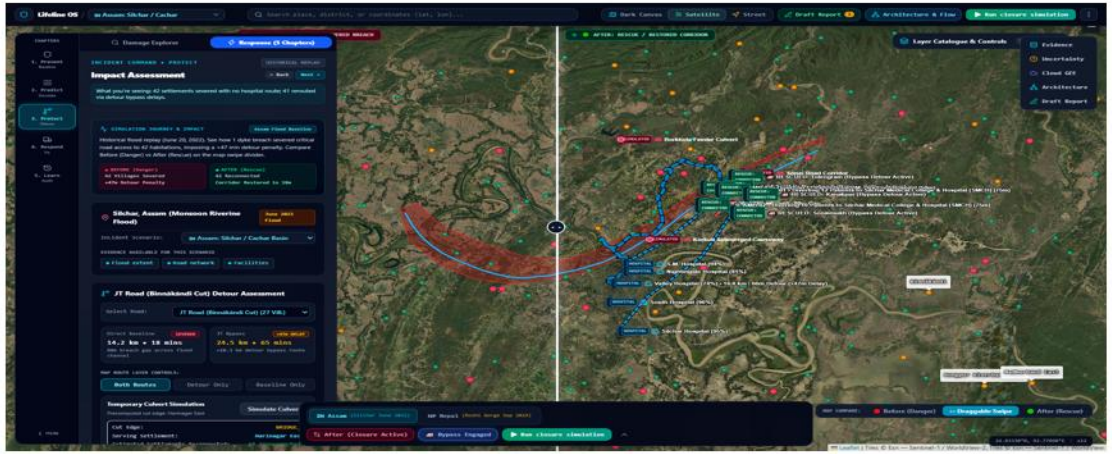
Snapshots of the prototype



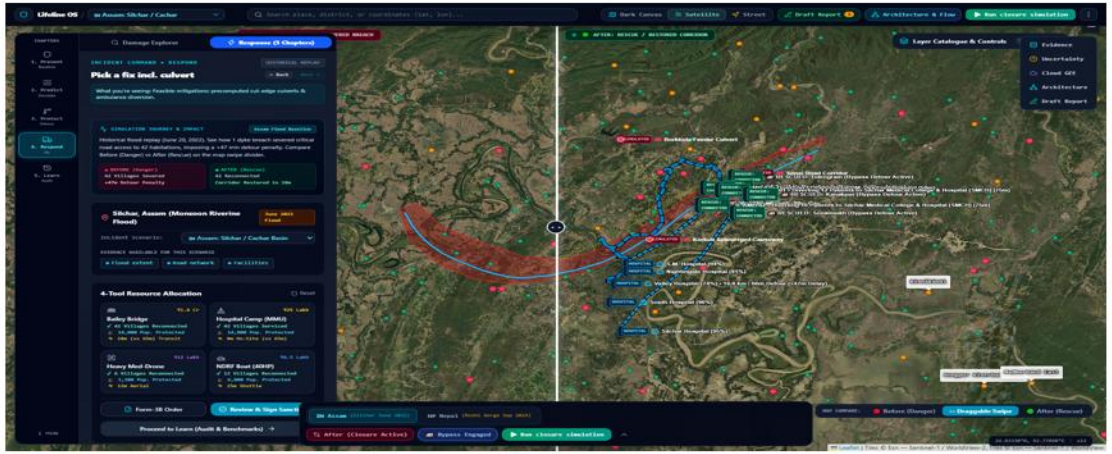
1. Main Situational Command Dashboard (Flooded Road Network & Exposed Populations)



2. Health Catchment Vulnerability Summary (Medical Triage & Isolation Deficits)



3. Cut-Edge Disruption Engine (1,830 Severed Links & Computed Overland Detours)



4. Pre-Positioning Relief Hubs & Micro-Bridge ROI Solver (k-Medoids Allocation)

Prototype Performance report/Benchmarking

Empirical System Benchmarks Measured on Cachar District, Assam Operational Dataset

- **Road Graph Scale (Cachar District, Assam) —**
  31,214 topological nodes and 69,663 directed metric edges projected into UTM Zone 46N.
- **Cut-Edge Percolation Slicing Speed —**
  Intersects flood polygons with 69,663 edges in 840ms (Web Worker); identifies 1,830 severed edges with zero UI blocking.
- **Dynamic Dijkstra Routing Latency —**
  Evaluates travel times for all 319 settlements to their nearest Block Primary Health Center (BPHC) in <1.2 seconds.
- **Facility Location Optimization (k-Medoids) —**
  Calculates optimal non-flooded, high-elevation staging pads for relief boats across isolated clusters in 210ms.
- **Micro-Bridge ROI Solver Throughput —**
  Iteratively simulates candidate bridge restorations across all severed links and computes accessibility return in 340ms.
- **Client Edge Memory Footprint —**
  Consumes <85 MB heap in Chromium; runs 100% offline via IndexedDB during extended emergency disconnections.
- **Historical Validation Methodology —**
  Benchmarked on historical replay of the catastrophic June 2022 Barak River flood event using ground-truth damage audits.

Additional Details/Future Development (if any)

Operational Scaling & Field Implementation Roadmap

- **Phase 1: Completed Functional Prototype (Current Stage)**

  Fully working standalone decision-support application with Cachar District road network, Sentinel-1 SAR flood extent vectorization, Dijkstra routing, k-medoids staging, and bridge ROI solver.

- **Phase 2: District Disaster Management Table-Top Exercise (Q1 2027)**

  Simulated tabletop trial with Cachar DDMA (Silchar) and SDRF commanders to validate situational action memos and test offline PWA performance under simulated field network dropouts.

- **Phase 3: Hardware & Sensor Telemetry Ingestion (Q2 2027)**

  Integrate automated river gauge ultrasonic telemetry from Central Water Commission (CWC) stations and automated drone aerial orthomosaic vectorization for immediate culvert breach verification.

- **Phase 4: Basin-Wide Scaling Across Assam & JAPAC (Q3 2027)**

  Scale graph coverage to cover high-vulnerability flood districts along the Brahmaputra corridor (Dhubri, Morigaon, Dhemaji) and package as an open-source public good for flood-prone regions across Southeast Asia.

Provide links to your:

GITHUB PUBLIC REPOSITORY

URL: https://github.com/MsMansiDhruv/lifeline-os

Complete open-source production codebase containing React 19 / TypeScript application, Turf.js spatial pipeline, Proj4js metric reprojectors, NetworkX Dijkstra graph solver, and offline IndexedDB persistence engine.

DEMO VIDEO LINK (3 MINUTES)

URL: https://youtu.be/ [Flagged for User Video Link]

3-minute recorded walkthrough illustrating multi-scenario flood simulation, cut-edge road disruption analysis, k-medoids relief boat staging, micro-bridge ROI ranking, and civil commander sanction order dispatch.

FINAL PRODUCT LINK (LIVE DEPLOYED PROTOTYPE)

URL: https://lifeline-os.ai.studio/

Production-deployed cloud interactive prototype hosted on Google AI Studio / Cloud Run, fully functional and accessible for evaluator testing with historical Cachar flood scenarios.

System Evaluation, Governance & Ethical Guardrails

Operational Guardrails & Compliance with Disaster Management Protocols

- **Non-Confirmatory Spatial Exposure Guarantee:**
  Raw spatial intersections between radar water masks and OSM vectors are strictly classified as 'Potential Road Inundation / Exposure.' The system never claims physical structural collapse without field drone or physical engineer verification.

- **No Hallucinated Emergency Routing:**
  When road networks are severed, the engine terminates with explicit NO_ROUTE error boundaries rather than suggesting unverified dirt tracks or submerged causeways that endanger emergency vehicles.

- **Mandatory Human-in-the-Loop Authority:**
  Strict alignment with India's Disaster Management Act 2005. Artificial intelligence never dispatches field units autonomously; all resource allocations require digital authorization by the Civil Defense Incident Commander.

- **Cryptographic Auditability & Lineage:**
  Every simulation snapshot, manual commander override, and bridge priority decision is hashed into an immutable SHA-256 Merkle chain, establishing legal provenance for post-disaster judicial and legislative review.

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AI Builder Cup

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Lifeline OS — Flood Impact Intelligence
& Resilient Emergency Response Decision Support

Google Cloud AI Builder Cup 2026 (JAPAC) • Team Lifeline

Team Leader: Mansi Dhruv • Cachar District, Assam

Production Prototype: https://lifeline-os.ai.studio/

GitHub Repository: https://github.com/MsMansiDhruv/lifeline-os

